



PHENOTYPIC AND GENOTYPIC CHARACTERIZATION OF METHICILLIN RESISTANT *Staphylococcus aureus* FROM FOMITES IN A TERTIARY HOSPITAL IN TACLOBAN CITY

Pete Andrei Arce
Hazel Conarco
Agripino Limpiado

*Dona Remedios Trinidad Romualdez Educational Foundation Inc
Tacloban City, Philippines*

<https://doi.org/10.5281/zenodo.11157655>

ABSTRACT

Methicillin-Resistant *Staphylococcus aureus* (MRSA) is one of the most common nosocomial bacterial pathogens which could be transmitted by health personnel themselves, non-invasive fomites and hospital environment. Due to its resistant phenotype, it is considered as one of the notorious pathogens causing wide range of infections which are difficult to treat due to limited antibiotic choices. Primary aim of the study is to determine the presence of MRSA among fomites in one of the tertiary hospitals in Tacloban City. Samples were collected from fomites using swab technique in two different areas namely: General areas (hospital wards) and Special Areas (NICU, ICU, OR-DR complex). *Staphylococcus aureus* was identified using the conventional method while phenotypic characterization of potential MRSA was done through Antimicrobial Susceptibility Testing using Oxacillin and Cefoxitin as recommended by CLSI, then followed by genotypic characterization using molecular technique. Of the 64 isolates, 57 (37 from general areas, 20 from special areas) were confirmed to be *S. aureus*. Of the 57, nineteen (19) or 51.4% from general areas were resistant to Oxacillin and Cefoxitin while seven (7) or 35% from special areas. Based on the phenotypic characterization, this indicates as potential MRSA. On the other hand, using PCR, the 19 potential MRSA from general areas were found to have *mecA* gene which means, all the isolates were MRSA. From the special areas, sixteen (16 of 20) or 80% of the isolates were true MRSA as indicated by the presence of *mecA* gene. Based on the genotypic characterization of isolates from special areas, a relatively high percentage of potential MRSA were not detected using the phenotypic characterization could be noted. Surface contamination of fomites within healthcare setting remains to be one of the challenges the public healthcare needs to overcome. Thus, the need to strictly implement decontamination techniques to

reduce the rate of such contamination is still apparent. Moreover, as what the study clearly illustrates, the need to perform confirmatory tests like PCR in identifying MRSA through molecular analysis is of utmost importance for proper microbial identification.

Key terms: MRSA, Fomites, PCR Phenotypic/ Genotypic characterization

INTRODUCTION

Background of the Study

Staphylococcus aureus remains one of the most virulent pathogens to humans. One reason that accounts for its virulence is its ability to cause a wide range of infections involving any organ across the body. It is notorious for causing boils, furuncles, impetigo and other superficial skin infection in humans. However, in more frequent cases, particularly involving patients in hospitals who are debilitated by chronic illnesses, it may also cause more serious nosocomial infections like pneumonia, deep abscesses, osteomyelitis, endocarditis, meningitis and in severe cases, septicemia. The success of *S. aureus* as a pathogen and its ability to cause such a wide range of infections are the result of its extensive virulence factors. The increase in the resistance of this virulent pathogen to antibacterial agents, coupled with its increasing prevalence as a nosocomial pathogen, is of major concern. The core resistance phenotype that seems to be most associated with the persistence of *S. aureus* in the hospital is methicillin resistance (Richmond, 2008).

Methicillin is a β -lactam antibiotic that is resistant to the action of β -lactamase secreted by many penicillin-resistant bacteria. It acts via competitive inhibition of transpeptidase enzyme by its affinity to penicillin-binding protein 2 (PBP2) used by bacteria to cross-link the peptide (D-alanyl-alanine) mandatory for peptidoglycan synthesis. It was developed to treat staphylococcal infections. β -lactam antibiotics, such as penicillin and its derivatives, have poor affinity for this altered PBP, and organisms are not killed when exposed to common therapeutic concentrations (Davies, J., & Davies, D. 2010).

Methicillin resistance in *S. aureus* is primarily mediated by the *mec A* gene, which codes for the modified penicillin-binding protein 2a (PBP 2a or PBP 2'). PBP2a is located in the bacterial cell wall and has a low binding affinity for β -lactams. Although all cells in a population of *S. aureus* may carry the *mec A* gene, only a few of the cells will express the gene, thus, both resistant and nonresistant bacteria can exist in the same culture. *Mec A* expression can be constitutive or inducible. *Mec A* is carried on a mobile genetic element, the staphylococcal cassette chromosome *mec*. The occurrence of PBP2a means MRSA is not only opposing to methicillin but, moreover to all β -lactam antibiotics together with synthetic penicillin, cephalosporins and carbapenems (Fishovitz, J., Hermoso, J. A., Chang, M., & Mobashery, S. 2014).

Antimicrobial resistance is now considered as a global health problem that increases the morbidity, mortality and financial cost of treating infectious diseases (Salam, et al, 2023). Increasing reports of antimicrobial resistance to various drugs and occurrence of methicillin resistant strains compound the problem. According to Antimicrobial Resistance Surveillance Program, Research Institute, for Tropical Medicine, and Department of Health, Philippines (DOH, 2016) there were 3,119 MRSA isolates reported from different hospital sites in the Philippines for 2016. The overall cumulative MRSA rate for 2016 was at 61.5% with variations by sentinel site MRSA rates ranging from as low as 19.9% to as high as 81.5%. Our country faces serious threat of antimicrobial resistance.

In healthcare setting, MRSA can be harder to treat due to its resistance to most commonly used antibiotics. It can cause surgical wound infection, pneumonia, or infections of catheters inserted into the veins. It may also include blood stream infection, heart valve infection, soft tissue infection, abscesses in organs, bone infection and joint infection.

Both phenotypic and genotypic markers can be used in identifying MRSA. In this study, the researchers employed the most common method of phenotyping and the latest trend in genotypic characterization using the PCR-based molecular technique through detection of *mecA* gene. The antibiotic susceptibility test was performed through the disk-diffusion method based on Clinical and Laboratory Standard Institute (CLSI) guidelines. The isolates were challenged with Cefoxitin and Oxacillin to test for its resistance. PCR detects the *mecA* gene and, in addition, a highly specific sequence for *S. aureus* by polymerase chain reaction (PCR) and reverse hybridization. Molecular genetic testing with the PCR needs much less time than conventional microbiology-based methods. Therefore, genetic testing provides not only a considerable advantage with respect to reliability but also to speed.

The challenge today is to keep on making bacterial databases linking genetic markers and clinical outcomes so that important correlations of the disease can be detected.

General Objective

This study aimed to investigate the phenotypic and genotypic characteristic of MRSA isolates obtained from fomites in participating Hospitals in Tacloban City, Philippines, using both AST and PCR.

Specific Objectives

1. To Isolate and characterize MRSA obtained from fomites of a tertiary hospital in Tacloban City as to:
 - 1.1. Susceptibility/Resistance to Oxacillin and Cefoxitin
 - 1.2. Presence of *mecA* gene through PCR
2. Determine the prevalence of MRSA among fomites in the following:

- 2.1. General Areas
- 2.2. Special Areas

METHODOLOGY

A descriptive cross-sectional design was utilized in the study. The study was conducted in one of the tertiary hospitals in Tacloban City. Different fomites from the identified sampling areas were sampled and processed for the isolation and identification of MRSA, then, the data collected were treated with appropriate statistical tools.

Sampling was preceded by the categorization of the sampling areas into two namely, General Areas and Special Areas. General areas include the four hospital wards- maternity ward, pediatric ward, surgery ward, and geriatric ward-while the special areas will include the NICU, ICU, ER and OR-DR complex. The researchers sampled fomites unique to each sub-area identified. A total of 39 samples were taken from the general area and a total of 25 samples from the special area. A total of 64 swab samples from bed railings, bed spreads, chairs, fan switches, sink faucet, door knob, light-plug-ins, pillow, drip stands, patient charts, sphygmomanometer stethoscope, tabletops, etc. were collected aseptically from all the areas of the hospital identified in the study.

Since the research focuses on phenotypic and genotypic characterization of suspected MRSA isolates, the data gathering procedure was divided into two major, interconnected tests for characterization- the AST and the PCR.

Direct swabbing from identified fomites was done for the isolation of potential MRSA. The researchers who conducted the sampling were equipped with personal protective equipment such as sterile gloves, hair net, face mask and laboratory gown to prevent contamination that could possibly interfere with the sampling process. The sterile cotton swab was aseptically moistened with distilled water. Swabbing was done on the following manner: the swab was held at a slight angle to the surface and moderate pressure was applied to allow as much contact between the swab tip and surface of the fomites, parallel strokes from left to right, and top to bottom of sampling area, rolling the swab approximately 180 degrees in a clockwise direction, ensuing the previously swabbed area is slightly overlapped. The same procedure was repeated vertically at 90 degrees and diagonally at 45 degrees. After sampling, the swab was immediately soaked in TSB, incubated at 37 degrees Celsius for 24 hours.

The bacterial growth on the surface of the TSB was inoculated into Mannitol Salt Agar enriched with 7.5% NaCl, pH at 7.2-7.4 for the isolation *S. aureus*. Equipped with PPE, the performing researcher aseptically inoculated bacterial isolates from TSB onto the MSA using the streak plate method. The inoculated samples were incubated at an inverted position under 37 degrees Celsius for 24 hours. Growth in MSA is an indication of staphylococcus species which then subjected to coagulase test to differentiate *S. aureus* from other species of Staphylococci.

Phenotypic Characterization using Antimicrobial Susceptibility Tasting

Disk Diffusion Method was used in the AST. All the procedures observed in this part of the method are based on the CLSI-2017-M100-S27 guidelines of standardized procedure for Antimicrobial Susceptibility Test for Methicillin Resistance of *Staphylococcus aureus*. The isolates' reaction to the oxacillin (1 µg) and cefoxitin (30 µg) disks was recorded and interpreted based on the breakpoints provided in the CLSI-2017-M100-S27. For the preparation of the inoculum, the performing researcher used the direct colony suspension method to make a suspension of the organism in saline to the density of a McFarland 0.5 turbidity standard, approximately corresponding to $1-2 \times 10^8$ CFU/mL for *S. aureus*. A sterile loop was used to pick colonies from an overnight culture on MSA. Several morphologically similar colonies were used to avoid selecting an atypical variant. The colonies were suspended in saline and mixed to an even turbidity. Adjustment of the density of the organism suspension to McFarland 0.5 was done by adding saline or more bacteria.

. A sterile cotton swab was dipped into the suspension, then spread evenly over the entire surface of Mueller Hinton Agar (MHA) ensuring that there are no gaps between streaks. Cefoxitin and Oxacillin disks were applied within 15 minutes of inoculation. The plates then incubated at 33-35 degrees Celsius for 24 hours.

. As provided by CLSI M100 document. Breakpoints for results will be observed as follows: ≤ 24 mm = *mecA* positive ≥ 25 mm = *mecA* negative. Results of AST were recorded. All isolates, regardless of reaction to AST were processed for PCR for molecular test of presence of *mecA* gene.

Genotypic Characterization using Polymerase Chain Reaction (PCR)

PCR amplification is considered as the golden standard in the identification of MRSA by detection of *mecA* gene. PCR is valuable because it is simple, sensitive, and powerful. PCR can amplify a single copy of a target DNA into an exponential amount of nucleic acid product over the course of 25 to 40 reaction cycles.

Heat Treatment was used in extraction of DNA as described by Ashti et. al. in 2009. Each inoculum was emulsified in a separate tube with 1 ml NSS. The emulsified bacteria were then boiled (99 degrees Celsius) for 15 minutes in water bath. Afterwards, the emulsified bacteria were cooled down and centrifuged at 1000 rpm for 2 minutes. Five microliters of the supernatant were used for PCR.

For this study, the researchers used the primer sequence (5'-3') TAGAAATGACTAACGTCCG with an amplicon of 154 bp at nucleotide positions 178-198 (Taweeporn, et. al., 2011). PCR was performed using a master mix containing 69 µl of sterile water, 10 µl of 10× amplification buffer, 10 µl of 2 mM deoxynucleoside triphosphates, 3 µl of 3.75 mM MgCl₂, 1 µl of a 10 pM stock of each primer, and 0.5 µl (2.5 U) of *Taq* polymerase. Amplification buffer (10×), deoxynucleoside triphosphates, MgCl₂, and *Taq* polymerase will be from the *Taq* PCR core kit (Qiagen, Inc., Valencia,

Calif.). Template DNA (1 µl) was added to a 0.5-ml thin-walled PCR tube, followed by the addition of 99 µl of PCR master mix. After mixing, the sample was pulse centrifuged for 5 s and then overlaid with 50 µl of mineral oil. Cycling parameters were determined based on the protocol provided by the manufacturer sheet. Positive control that will be used is *S. aureus* 50A247.

Aliquots (5 µl) of the amplification products were analyzed by agarose gel electrophoresis using 2.0% LE agarose (FMC Bioproducts, Rockland, Maine) containing 0.5 µg of ethidium bromide per ml. Aliquots were run in parallel with a 100bp Ladder molecular weight marker on a 2% agarose gel for 1h at about 130V. Gel was stained in Ethidium bromide circa for 20-30min. It was de-stained briefly in milliQ water. Gels were visualized and photographed using a GDS7500 gel documentation system (UVP Inc., Upland, Calif.).

RESULTS AND DISCUSSION

The researchers studied 37 *S. aureus* isolates recovered from fomites within the general area and 20 within the special area for a total of 57 *S. aureus* isolates. Of the 57 *S. aureus* isolates tested, 35 were determined to be MRSA and 19 through the detection of *mecA* gene by PCR. The results of the Antimicrobial Susceptibility Test using Oxacillin (1 ug) and Cefoxitin (30 ug), contrasted with the PCR results, for the isolates from the general area (Table 1) and special area (Table 2) are shown below.

Table 1.
Results of Phenotypic Tests and Genotypic Tests performed with the isolates recovered from the general area in the hospital.

General Areas	Number of Isolates (n)	Phenotypic (susceptibility to Oxacillin/Cefoxitin)	Genotypic (Presence of <i>mecA</i> gene)
North Wing	9	3 (33.3%)	5 (55.5%)
South Wing	11	7 (63.6%)	7 (63.6%)
West Wing	17	9 (52.9%)	7 (41.2%)
TOTAL	37	19 (51.4%)	19 (51.4%)

Of the 37 *S. aureus* isolates recovered from the general area, 19 were found to be *mecA* gene positive therefore identified as **MRSA** and 18 were *mecA* gene negative therefore identified as **MSSA**. Disagreements with the results between AST and PCR were observed, however, Oxacillin and Cefoxitin Disk Diffusion Tests were consistent with PCR in 16 out of 37 isolates from the general area.

Table 2.
Results of the Phenotypic Tests and Genotypic Tests performed with the isolates recovered from the special areas in the hospital.

Special Areas	Number of Isolates (n)	Phenotypic (susceptibility to Oxacillin/Cefoxitin)	Genotypic (Presence of <i>mecA</i> gene)
OR	4	2 (50.0%)	4 (100%)
RR	4	0	3 (75.0%)
DR	5	3 (60.0%)	5 (100%)
NICU	4	1 (25.0%)	1 (75.0%)
ICU	3	1(33.6%)	3 (100%)
TOTAL	20	7 (35.0%)	16 (80.0%)

Of the 20 *S. aureus* isolates recovered from the special area, 16 were found to be *mecA* gene positive therefore identified as **MRSA** and 4 were *mecA* gene negative therefore identified as **MSSA**. Disagreements with the results between AST and PCR were observed, however, Oxacillin and Cefoxitin Disk Diffusion Tests were consistent with PCR in 8 out of 20 isolates from the special area.

An important objective established at the beginning of this research is the determination of the amount of contamination in fomites within the hospital. Several different fomites were sampled and tested first for microbiological identification and then proceeded with the phenotypic characterization (AST) and genotypic characterization (PCR) once they were confirmed as *S. aureus* isolates. However, the researchers' main fomite of interest were the patient charts from the nurses' station from all wards within the hospital. This is because these fomites come in contact more frequently with the hospital staff like nurses and doctors, and oftentimes, with patients. Other fomites where *S. aureus* isolates were recovered from included table tops, poster frame, keyboards, bed cases, light switch, door knobs, Stirrup, white board, Kelly's pad, bed railings, thermometer, sphygmomanometer, breathing apparatus, and stethoscope.

DISCUSSION

To this day, nosocomial pathogens continue to cause complications to debilitated patients in hospitals. Although, infection control measures are implemented in different hospitals to manage the prevalence of nosocomial infections, the existence of these pathogens in the hospital setting is still apparent. MRSA is one of the most common nosocomial pathogens in hospitals. This is less surprising compared to other known nosocomial pathogens given that MRSA originating from outside healthcare institutions has been very common in the last 10 years with the rapid clonal expansion of community-associated MRSA. Often times, strains of MRSA recovered from within the hospitals are brought in from communities by different patients and visitors coming from different places. Nonetheless, once these strains enter the hospital setting, patients are always at risk especially those whose immunity is compromised. Many entities in the hospital could

harbor nosocomial pathogens such as MRSA. Even hospital personnel and staff could harbor such pathogen and remain without symptoms. Oftentimes, they could even be unexpected vectors of the pathogen, possibly transmitting them to patients as they have direct interaction with them. But recent studies on the prevalence of MRSA in different hospital fomites have shown that fomites are more effective vectors in transmitting the pathogen to patients exposed to such fomites. Because of this, the researchers have deemed it necessary to investigate the presence of MRSA in fomites found within the hospitals in the locale, the city of Tacloban.

A total of 64 fomites from 2 different areas in the hospital namely, general area and special area were sampled and processed for identification of MRSA. Conventional methods for identification of *S. aureus* were employed to properly identify the isolates recovered from the fomites. After growth in TSB, the isolates were sub-cultured in MSA. MSA is both a selective and differential medium for the isolation of *S. aureus*. MSA allows growth of both coagulase negative and coagulase positive *Staphylococcus* species in the medium, however the phenol red component in MSA makes it a selective medium between CoNS and CoPS since of the two species, only *S. aureus* ferment mannitol. The 7.5% NaCl component of the medium is nearly 10 times the usual concentration seen in most media. This high salt concentration inhibits growth of other non-halophilic microorganisms especially in mixed flora specimens. Apart from the medium used, special conditions were observed for optimal recovery of *S. aureus*. Cultures were incubated for 24 hours at 37 degrees Celsius. Stacking of petri dishes were minimized for optimal distribution of temperature among the plates. A minimum of 3 petri dishes were stacked together in the incubator for growth within 24 hours. After exactly 24 hours, growth on all plates was observed, however, out of 64 isolates recovered from the fomites, only 57 were confirmed as *S. aureus* following the tube coagulase test. For the tube coagulase test, human plasma was used instead of bovine plasma. There was no particular reason for this preference.

All 57 confirmed *S. aureus* isolates were processed for identification of MRSA using both phenotypic and genotypic tests. Standard procedure in performing Antimicrobial Susceptibility Test was employed, however, for this study, the researchers decided to enrich the MHA with 2% NaCl to optimally induce the phenotypic expression of the *mecA* gene. This enhancement of the medium has been shown to increase the expression of the *mecA* gene in some cases of strains reported to have suppressed expression of the *mecA* gene. Both oxacillin and cefoxitin were used in the AST.

PCR remains as the gold standard for the identification of MRSA through the detection of the *mecA* gene which is responsible for the resistant phenotype of MRSA. An essential part of PCR is the extraction of DNA since this phase of the process practically ensures that there will be an analyte to be detected in the first place. Usually, in laboratory settings, especially in higher institutions, DNA extraction is done through the use of a kit which uses spin columns containing a silica resin that selectively binds DNA (or RNA). This technique ensures optimal recovery of DNA. However, in this study, the researchers employed the boiling method which includes boiling 200 μ L of the bacterial suspension for 15 minutes at 99 degrees Celsius. This technique has been proven to

have almost the same yield of DNA as the silica resin technique. In this study, to make sure that negative results in PCR aren't due to the complete absence of DNA because of a failed extraction, the researchers combined the extracted DNA with 16s RNA so that even if the *mecA* gene is absent in the isolate, 16s RNA will be visualized after gel electrophoresis. This will help the researchers make sure that the DNA extraction phase of PCR did not fail and the absence of band in the characteristic region is due to the absence of *mecA* gene and not the complete absence of DNA in the solution. Double bands were observed in some samples in the visualization of results following gel electrophoresis. Samples 1, 3, 4, 5, 6, and 7 (Figure 2) showed the characteristic 162 bp region of *mecA* in 1% agarose gel. The rest of the samples positive for *mecA* gene visualized bands at the 310 bp region with others showing variability and double bands. Double bands and variability in the location of the characteristic band for *mecA* gene were observed in samples 18, 22, 23, 24, 28, 32, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 48, 49, 54, 56, 58, 59, and 60 (Figure 3). All of these samples were interpreted as positive for *mecA* gene.

In this study, all 57 isolates were processed through PCR regardless of their reaction in AST. This is due to the researchers' anticipation of the emergence of strains whose resistant phenotype is not expressed despite having the *mecA* gene and vice versa. True enough, disagreements between The AST and PCR were observed. Although disagreements between the AST and PCR are a common story in microbiology, in this study, the number of disagreements between AST and PCR prompted the researchers to further analyze the data. These disagreements led the researchers to compute for the percentage agreements between methods and the sensitivity, specificity, positive predictive value and negative predictive value of the methods employed in this study, putting into consideration that these figures will only serve to analyze the methods used in this study as they were carried out and not of the integrity of the methods themselves as used in other clinical laboratories.

Among the 57 isolates, 39 were found to be MRSA through PCR. 24 of these were correctly identified as MRSA by combined ODD and CDD methods while correctly detecting 20 MSSA isolates. This leaves AST with a sensitivity of 92.3% and a specificity of 64.5%. Rapid and precise identification of MRSA is of utmost importance in the clinical microbiology laboratories for institution of appropriate treatment. However, there is no optimal phenotypic method for detection of methicillin resistance in *S. aureus* as conventional methods require special conditions e.g. 2% NaCl enriched media, up to 48 hours of incubation time, $\leq 35^{\circ}\text{C}$ temperature, etc. Despite necessary precautions and expertise in handling MRSA, discrepancies occur as far as phenotypic methods are concerned. Anand et al., 2009 found cefoxitin disc method to be 100% in concordance with PCR while in some cases discrepancies have been found even while using cefoxitin disc for MRSA detection, PCR based genotypic methods, although often referred to as a gold standard are not able to detect the non-*mecA* mediated resistances. Even if an isolate possesses *mecA* gene and positive in PCR, it does not necessarily imply that the strain is phenotypically MRSA, as there are occasions where the gene product is not expressed or the gene expression is suppressed.

Conclusions

In conclusion, the study described the prevalence of MRSA contamination on surfaces of hospital fomites and their distribution within the hospital. The over-all point prevalence of MRSA in the fomites in the Tertiary Hospital in Tacloban City was 61.4%, with the highest number of MRSA isolated from fomites were seen in the General Area of the hospital with 54.3% of the total number of MRSA detected through PCR with most of the number coming from its South and West Wings. MRSA was recovered in all rooms sampled but was seen least in the Special Areas of the Tertiary Hospital. The NICU recorded the lowest number of MRSA recovered. It is evident that contamination of hospital fomites remains as an existing threat to patients and hospital staff alike and that stricter compliance to infection control measures and antibiogram implementation are needed.

Recommendations

It is recommended, if possible clinical laboratories should use PCR in confirming potential MRSA isolates from the patient for proper antibiotic management and therapy. The researchers also suggest the use of automated molecular techniques in identifying strains of *S. aureus* with non-mecA mediated resistance such as PBP2a- latex agglutination test, Vitek 2 automated system and even Southern blotting and PFGEA. Likewise, a periodic nasal swab for Doctors and Nurses and other healthcare providers in hospital sitting is also highly recommended.

Compliance with Ethical Standards

Ethics is an integral part of conducting research, especially when the proponents are dealing with the people, the community, pertinent documents, animals, and the like. An absolute observance was done and followed by the ethical consideration in conducting this study to avoid fallacy or misconceptions of the participant's views about the phenomenon under study, its integrity, and most especially to protect them from any harm by hiding their personal information using John Doe names. Lastly, the proponent asked the approval of the consent form from the participants, stating their rights and privileges including their role in the research.

Acknowledgments

The researchers would like to extend their gratitude to their families who supported them, and above all, to the greatest provider of all, Our Lord Almighty. To God be the glory.

REFERENCES

- Davies, J., & Davies, D. (2010). Origins and evolution of antibiotic resistance. *Microbiology and molecular biology reviews : MMBR*, 74(3), 417–433. <https://doi.org/10.1128/MMBR.00016-10>
- Fishovitz, J., Hermoso, J. A., Chang, M., & Mobashery, S. (2014). Penicillin-binding protein 2a of methicillin-resistant *Staphylococcus aureus*. *IUBMB life*, 66(8), 572–577. <https://doi.org/10.1002/iub.1289>
- Richmond, T. (2008). *Staphylococcus*. Medical Microbiology - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK8448/>
- Salam, M. A., Al-Amin, M. Y., Salam, M. T., Pawar, J. S., Akhter, N., Rabaan, A. A., & Alqumber, M. A. A. (2023). Antimicrobial Resistance: A Growing Serious Threat for Global Public Health. *Healthcare (Basel, Switzerland)*, 11(13), 1946. <https://doi.org/10.3390/healthcare11131946>